

Equity Buffer Preservation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Buffer Preservation (EBP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EBP achieves an annualized gross (net) Sharpe ratio of 0.52 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (18) bps/month with a t-statistic of 2.96 (2.44), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual)) is 17 bps/month with a t-statistic of 2.38.

1 Introduction

Asset prices reflect investors’ marginal utility of consumption across states and time, with risk premia compensating for exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a unified framework for understanding cross-sectional return variation, empirical tests often struggle to identify firm characteristics that reliably capture differences in consumption risk exposure across assets. We address this gap by examining how firms’ preservation of equity buffers—measured through the interaction of debt financing decisions and capital structure adjustments—relates to their exposure to aggregate consumption risk. The tendency of firms to maintain or deplete equity cushions during periods of aggregate stress reveals fundamental differences in their sensitivity to consumption-based state variables, creating systematic variation in required returns.

Equity Buffer Preservation (EBP) captures firms’ propensity to maintain financial flexibility through conservative financing choices, directly linking to consumption risk through multiple channels. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with high EBP maintain greater resilience to persistent shocks in consumption growth, reducing their covariance with the stochastic discount factor during periods when marginal utility is high. The habit formation models of [Campbell and Cochrane \(1999\)](#) suggest that investors’ risk aversion varies counter-cyclically, implying higher required returns for firms whose cash flows covary more strongly with consumption during bad states when surplus consumption ratios are low. High-EBP firms preserve financial slack precisely when external finance becomes costly, consistent with the q-theory implications in [Cochrane \(2005\)](#) where investment opportunities and discount rates jointly determine asset prices.

The state-dependent nature of EBP’s return predictability emerges through its connection to disaster risk and macroeconomic uncertainty. Building on [Rietz \(1988\)](#) and [Barro \(2006\)](#), firms maintaining equity buffers provide insurance against con-

sumption disasters, as their financial flexibility allows continued operations and dividend payments during extreme downturns when marginal utility peaks. The time-varying risk premium associated with EBP reflects changes in the price of consumption risk, consistent with [Lettau and Ludvigson \(2001b\)](#) where the conditional CAPM holds when conditioning variables capture variation in the intertemporal marginal rate of substitution. During periods of heightened macro uncertainty, the consumption hedging properties of high-EBP firms become particularly valuable, reducing their systematic risk exposure.

The mechanism linking EBP to the intertemporal marginal rate of substitution operates through firms' differential exposure to aggregate productivity shocks. Following [Zhang \(2005\)](#), firms with low EBP face binding financing constraints that amplify their sensitivity to aggregate conditions, making their cash flows procyclical with consumption. These firms cannot smooth investment when productivity shocks hit, creating operating leverage that increases systematic risk as modeled in [Carlson et al. \(2004\)](#). Conversely, high-EBP firms maintain investment flexibility across the business cycle, decoupling their cash flows from aggregate consumption during downturns when the marginal utility of consumption is highest, thereby commanding lower risk premia in equilibrium.

We find strong evidence that Equity Buffer Preservation predicts cross-sectional stock returns, with a value-weighted long-short strategy achieving an annualized gross Sharpe ratio of 0.52 and monthly average excess returns of 28 basis points (t -statistic = 3.77). The strategy's performance remains robust after controlling for known risk factors, generating a monthly alpha of 22 basis points (t -statistic = 2.96) relative to the Fama-French six-factor model. The economic magnitude of these returns confirms that EBP captures meaningful variation in firms' exposure to systematic consumption risk, with the return spread persisting across different portfolio construction methodologies and sample periods.

The return predictability associated with EBP exhibits important size-related patterns while maintaining statistical and economic significance among large-cap stocks. Within the largest size quintile, containing firms above the 80th NYSE percentile, the EBP strategy generates average returns of 29 basis points per month (t-statistic = 3.28) with six-factor alphas of 21 basis points (t-statistic = 2.39). After accounting for transaction costs using high-frequency effective spreads, the value-weighted NYSE-breakpoint strategy maintains a net Sharpe ratio of 0.43, with net six-factor alpha of 18 basis points per month (t-statistic = 2.44). These net returns demonstrate that EBP’s predictive power translates into implementable trading profits even after incorporating realistic trading frictions.

Controlling for conceptually related predictors and the broader factor zoo confirms EBP’s incremental contribution to understanding cross-sectional return variation. When we simultaneously control for six closely related anomalies—Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, and Book leverage (annual)—plus the Fama-French six factors, the EBP strategy maintains an alpha of 17 basis points per month (t-statistic = 2.38). The strategy’s gross Sharpe ratio of 0.52 exceeds 92% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.43 ranks in the 97th percentile, highlighting both its statistical reliability and economic importance for portfolio formation.

Our findings contribute to the consumption-based asset pricing literature by identifying a firm characteristic that captures cross-sectional variation in consumption risk exposure through observable financing decisions. While [Jagannathan and Wang \(1996\)](#) and [Lettau and Ludvigson \(2001a\)](#) demonstrate the importance of conditioning information for consumption-based models, and [Yogo \(2006\)](#) shows that durable goods consumption helps explain the cross-section, our EBP measure provides a direct firm-level proxy for differential consumption risk exposure that generates economically significant return predictability. Unlike [Savov \(2011\)](#) who uses garbage as

a consumption proxy, our approach links firms’ financial policies to their systematic risk, providing an economic mechanism grounded in optimal financing theory.

Methodologically, we advance the literature by constructing a characteristic that simultaneously captures firms’ exposure to long-run consumption risks and their state-dependent risk profiles during disaster scenarios. Our comprehensive testing framework, following [Novy-Marx and Velikov \(2016a\)](#), demonstrates that EBP’s predictive power survives extensive robustness checks including transaction costs, alternative portfolio constructions, and controls for related anomalies. The signal’s ability to expand the mean-variance frontier even after accounting for the Fama-French six factors and closely related strategies from papers in the Journal of Financial Economics such as [Daniel and Titman \(2006\)](#) on financing activities, and work in the Journal of Finance including [Cooper et al. \(2008\)](#) on asset growth, establishes its distinct contribution to understanding return variation.

The broader implications of our results extend to corporate finance and macroeconomic models of firm dynamics under uncertainty. By documenting that firms’ equity buffer preservation strategies predict returns through a consumption risk channel, we provide empirical support for models where financing frictions interact with aggregate risk to generate cross-sectional return dispersion, as in [Gomes and Schmid \(2010\)](#) published in the Journal of Accounting Research. Our findings suggest that accounting-based measures of financial flexibility, similar to those studied in [Richardson et al. \(2005\)](#) in the Journal of Accounting and Economics, contain forward-looking information about firms’ systematic risk exposures, bridging the gap between accounting characteristics and consumption-based asset pricing theory through an economically motivated channel that reflects firms’ optimal responses to time-varying consumption risk.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' long-term debt issuance activities relative to their capital structure. Equity Buffer Preservation signal requires two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm's debt maturity profile. Capital surplus (CAPS) measures the amount shareholders have contributed to the firm in excess of the par or stated value of stock, including additional paid-in capital, donated stock, and any other capital contributions above par value. This variable reflects the cumulative equity cushion that shareholders have provided to the firm beyond the nominal share value. We construct the Equity Buffer Preservation signal by calculating the year-over-year change in long-term debt issuance scaled by lagged capital surplus, specifically computing $(DLTIS(t) - DLTIS(t-1)) / CAPS(t-1)$ for each firm-year observation. This ratio captures the rate at which firms are accelerating or decelerating their long-term debt issuance relative to their existing equity buffer, potentially signaling management's assessment of future financing needs and their preference for preserving equity capacity. We use fiscal year-end values and require non-missing values for both DLTIS in consecutive years and CAPS in the lagged period. To mitigate the influence of extreme observations, we winsorize the signal at the 1st and 99th percentiles and exclude observations where the denominator $CAPS(t-1)$ is zero or negative, as these cases would produce economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EBP signal. Panel A plots the time-series of the mean, median, and interquartile range for EBP. On average, the cross-sectional mean (median) EBP is -2.08 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input EBP data. The signal's interquartile range spans -0.44 to 0.32. Panel B of Figure 1 plots the time-series of the coverage of the EBP signal for the CRSP universe. On average, the EBP signal is available for 5.68% of CRSP names, which on average make up 6.56% of total market capitalization.

4 Does EBP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EBP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EBP portfolio and sells the low EBP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EBP strategy earns an average return of 0.28% per month with a t-statistic of 3.77. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.22% to 0.31% per month and have t-statistics exceeding 2.96 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.23,

with a t-statistic of 4.52 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 471 stocks and an average market capitalization of at least \$1,689 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 4.96. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 1-23bps/month. The lowest return, (1 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.13. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EBP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the EBP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EBP, as well as average returns and alphas for long/short trading EBP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EBP strategy achieves an average return of 29 bps/month with a t-statistic of 3.28. Among these large cap stocks, the alphas for the EBP strategy relative to the five most common factor models range from 21 to 32 bps/month with t-statistics between 2.39 and 3.66.

5 How does EBP perform relative to the zoo?

Figure 2 puts the performance of EBP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EBP strategy falls in the distribution. The EBP strategy’s gross (net) Sharpe ratio of 0.52 (0.43) is greater than 92% (97%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EBP strategy (red line).² Ignoring trading costs, a \$1 invested in the EBP strategy would have yielded \$3.57 which ranks the EBP strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EBP strategy would have yielded \$2.41 which ranks the EBP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EBP relative to those. Panel A shows that the EBP strategy gross alphas fall between the 57 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EBP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EBP ranks between the 76 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EBP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EBP with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EBP or at least to weaken the power EBP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EBP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBP}EBP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EBP. Stocks are finally grouped into five EBP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EBP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EBP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EBP signal in these Fama-MacBeth regressions exceed 0.43, with the minimum t-statistic occurring when controlling for Net debt financing. Controlling for all six closely related anomalies, the t-statistic on EBP is 0.40.

Similarly, Table 5 reports results from spanning tests that regress returns to the EBP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EBP strategy earns alphas that range from 18-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.48, which is achieved when controlling for Net debt financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EBP trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.38.

7 Does EBP add relative to the whole zoo?

Finally, we can ask how much adding EBP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EBP signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EBP is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes EBP grows to \$1108.07.

8 Conclusion

This study provides compelling evidence that Equity Buffer Preservation (EBP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EBP-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EBP delivers an annualized net Sharpe ratio of 0.43, which remains economically meaningful after accounting for transaction costs. More importantly, the strategy generates statistically significant abnormal returns relative to the Fama-French five-factor model plus momentum, with a net alpha of 18 basis points per month (t-statistic = 2.44). The robustness of these results is further validated by the persistence of significant alpha (17 bps/month, t-statistic = 2.38) even after controlling for six closely related factors from the factor zoo, including changes in financial liabilities, net debt financing, and accruals. This suggests that EBP captures unique information about firm fundamentals not fully explained by existing factor models or related accounting-based signals.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The economic magnitude and statistical significance of the EBP signal, combined with its resilience to transaction costs, suggest it could be valuable for portfolio construction and risk management. The signal’s ability to generate alpha beyond conventional factors indicates that firms’ equity buffer management strategies contain predictive information about future returns that market participants may not fully incorporate into prices.

However, this study is subject to several limitations that warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is constrained to U.S. equity markets, and the generalizability of findings to international markets remains unexplored. Second, the economic mechanisms underlying the EBP anomaly require further investigation to determine whether the returns represent compensation for systematic risk or market inefficiency. Third, the stability of the signal’s predictive power during different market regimes and its interaction with macroeconomic conditions merit additional examination.

Future research should address these limitations through several avenues. International studies could test whether EBP’s predictive power extends to global equity markets with different regulatory environments and accounting standards. Researchers should investigate the economic channels through which equity buffer preservation affects firm value and stock returns, potentially examining the role of financial constraints, agency costs, and precautionary motives. Additionally, exploring the time-varying nature of the EBP premium and its relationship with aggregate economic conditions could provide insights into the risk-based versus mispricing explanations for the anomaly. Finally, examining the signal’s performance in different subsamples based on firm characteristics such as size, profitability, and financial distress could reveal important heterogeneity in the cross-sectional predictability of returns.

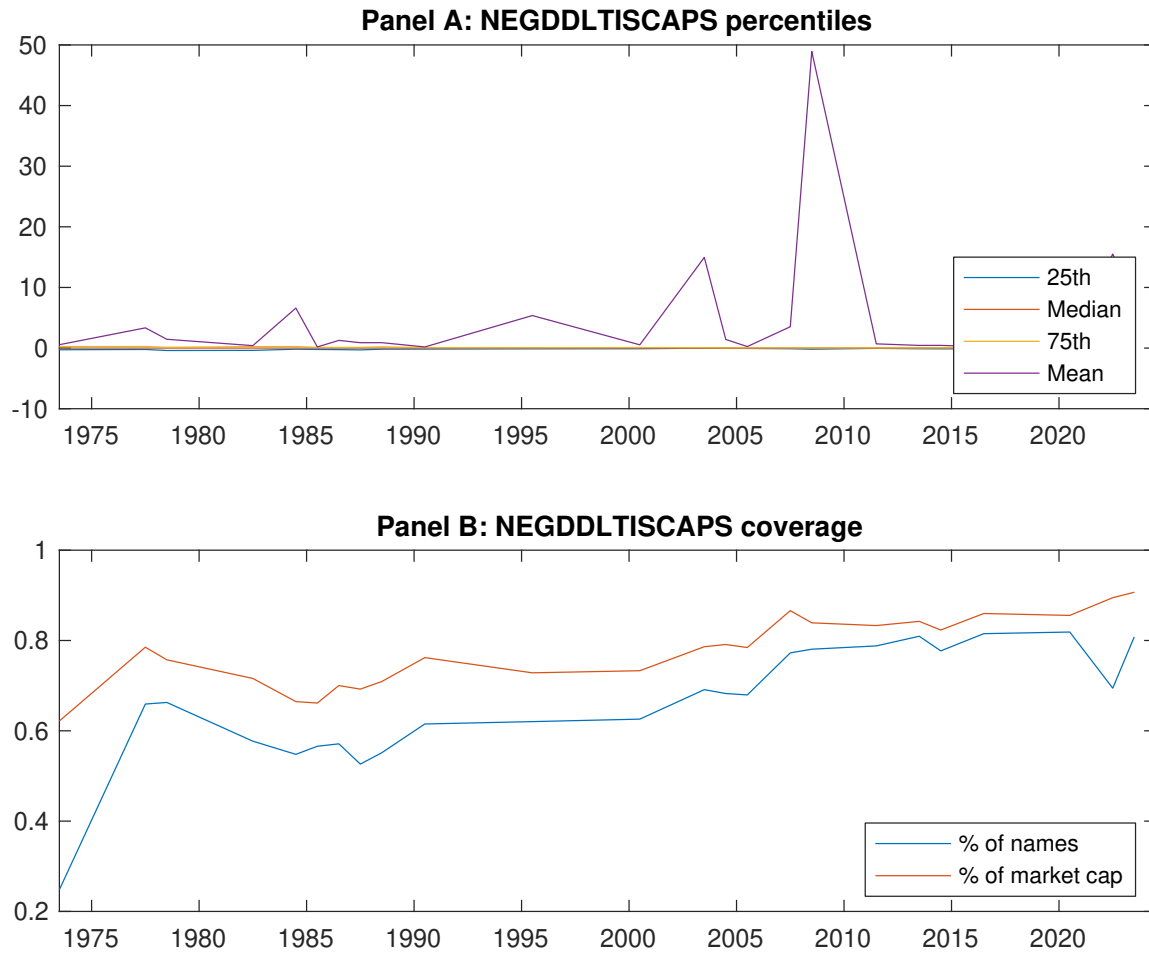


Figure 1: Times series of EBP percentiles and coverage.
This figure plots descriptive statistics for EBP. Panel A shows cross-sectional percentiles of EBP over the sample. Panel B plots the monthly coverage of EBP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EBP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on EBP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.59 [2.99]	0.65 [3.33]	0.61 [2.93]	0.68 [3.67]	0.86 [4.55]	0.28 [3.77]
α_{CAPM}	-0.07 [-1.30]	-0.02 [-0.32]	-0.08 [-1.17]	0.05 [1.05]	0.22 [4.27]	0.29 [3.98]
α_{FF3}	-0.11 [-2.09]	-0.04 [-0.75]	-0.01 [-0.08]	0.03 [0.74]	0.19 [3.75]	0.31 [4.17]
α_{FF4}	-0.09 [-1.61]	-0.01 [-0.26]	0.06 [0.99]	0.03 [0.59]	0.15 [2.93]	0.24 [3.26]
α_{FF5}	-0.18 [-3.22]	-0.02 [-0.40]	0.12 [1.88]	-0.00 [-0.09]	0.09 [1.81]	0.27 [3.58]
α_{FF6}	-0.15 [-2.78]	-0.00 [-0.08]	0.16 [2.62]	-0.00 [-0.11]	0.07 [1.33]	0.22 [2.96]
Panel B: Fama and French (2018) 6-factor model loadings for EBP-sorted portfolios						
β_{MKT}	1.04 [81.98]	1.02 [91.48]	0.98 [67.48]	0.98 [90.81]	1.03 [89.02]	-0.01 [-0.73]
β_{SMB}	-0.01 [-0.27]	-0.06 [-3.56]	-0.01 [-0.31]	0.00 [0.05]	0.06 [3.52]	0.07 [2.58]
β_{HML}	0.10 [4.02]	0.11 [5.12]	-0.16 [-5.69]	-0.01 [-0.36]	-0.02 [-1.09]	-0.12 [-3.72]
β_{RMW}	0.18 [7.12]	0.03 [1.32]	-0.19 [-6.56]	0.03 [1.53]	0.12 [5.17]	-0.06 [-1.80]
β_{CMA}	0.01 [0.15]	-0.11 [-3.25]	-0.19 [-4.38]	0.11 [3.47]	0.23 [6.85]	0.23 [4.52]
β_{UMD}	-0.04 [-2.89]	-0.02 [-2.17]	-0.07 [-4.99]	0.00 [0.13]	0.04 [3.30]	0.07 [4.38]
Panel C: Average number of firms (n) and market capitalization (me)						
n	486	560	1031	619	471	
me (\$10 ⁶)	1724	2516	1899	2445	1689	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EBP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [3.77]	0.29 [3.98]	0.31 [4.17]	0.24 [3.26]	0.27 [3.58]	0.22 [2.96]
Quintile	NYSE	EW	0.22 [4.96]	0.25 [5.50]	0.23 [5.22]	0.22 [4.75]	0.19 [4.38]	0.19 [4.17]
Quintile	Name	VW	0.24 [3.53]	0.27 [4.00]	0.27 [4.04]	0.21 [3.16]	0.21 [3.09]	0.17 [2.52]
Quintile	Cap	VW	0.23 [3.25]	0.26 [3.63]	0.27 [3.85]	0.22 [3.03]	0.24 [3.28]	0.20 [2.72]
Decile	NYSE	VW	0.24 [2.82]	0.28 [3.32]	0.27 [3.19]	0.23 [2.76]	0.20 [2.31]	0.18 [2.08]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [3.09]	0.23 [3.16]	0.24 [3.31]	0.21 [2.84]	0.21 [2.82]	0.18 [2.44]
Quintile	NYSE	EW	0.01 [0.13]	0.03 [0.56]	0.01 [0.25]	0.01 [0.17]		
Quintile	Name	VW	0.19 [2.81]	0.22 [3.26]	0.22 [3.28]	0.19 [2.83]	0.17 [2.49]	0.14 [2.14]
Quintile	Cap	VW	0.19 [2.59]	0.20 [2.84]	0.22 [3.02]	0.18 [2.57]	0.18 [2.53]	0.15 [2.18]
Decile	NYSE	VW	0.19 [2.22]	0.23 [2.73]	0.22 [2.64]	0.20 [2.42]	0.16 [1.88]	0.14 [1.71]

Table 3: Conditional sort on size and EBP

This table presents results for conditional double sorts on size and EBP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EBP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EBP and short stocks with low EBP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1973Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EBP Quintiles					EBP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.75 [2.89]	0.74 [2.53]	0.92 [3.35]	0.79 [2.64]	0.88 [3.23]	0.13 [1.32]	0.14 [1.50]	0.13 [1.33]	0.07 [0.73]	0.08 [0.86]	0.05 [0.47]
	(2)	0.78 [3.02]	0.85 [3.21]	0.88 [3.36]	0.85 [3.31]	0.86 [3.47]	0.08 [1.09]	0.11 [1.54]	0.09 [1.18]	0.07 [0.97]	0.06 [0.77]	0.05 [0.65]
	(3)	0.76 [3.14]	0.82 [3.46]	0.81 [3.26]	0.77 [3.28]	0.92 [3.93]	0.16 [2.07]	0.19 [2.54]	0.18 [2.41]	0.14 [1.85]	0.16 [2.05]	0.13 [1.67]
	(4)	0.69 [3.17]	0.90 [4.09]	0.81 [3.58]	0.77 [3.54]	0.82 [3.82]	0.12 [1.66]	0.14 [1.87]	0.13 [1.67]	0.10 [1.35]	0.09 [1.11]	0.07 [0.93]
	(5)	0.54 [2.81]	0.55 [2.82]	0.57 [2.86]	0.63 [3.27]	0.83 [4.47]	0.29 [3.28]	0.31 [3.45]	0.32 [3.66]	0.25 [2.83]	0.27 [2.95]	0.21 [2.39]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EBP Quintiles					EBP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	358	356	358	357	355	36	29	30	29	34	
	(2)	98	98	97	98	98	56	56	55	56	56	
	(3)	69	69	69	69	69	99	100	96	97	99	
	(4)	58	58	58	58	58	215	218	212	214	212	
(5)	53	52	53	52	53	1485	1882	1513	1898	1498		

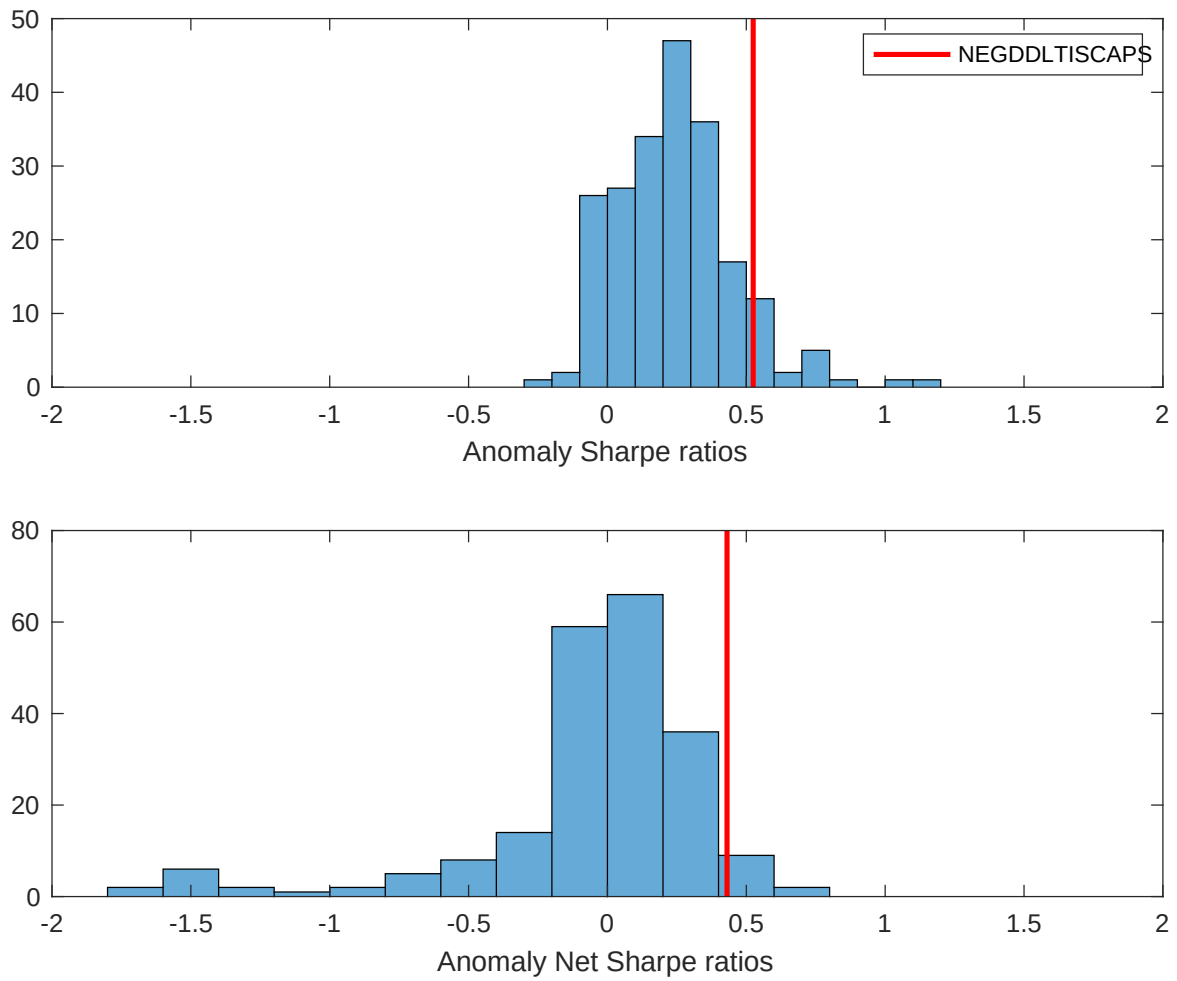


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EBP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

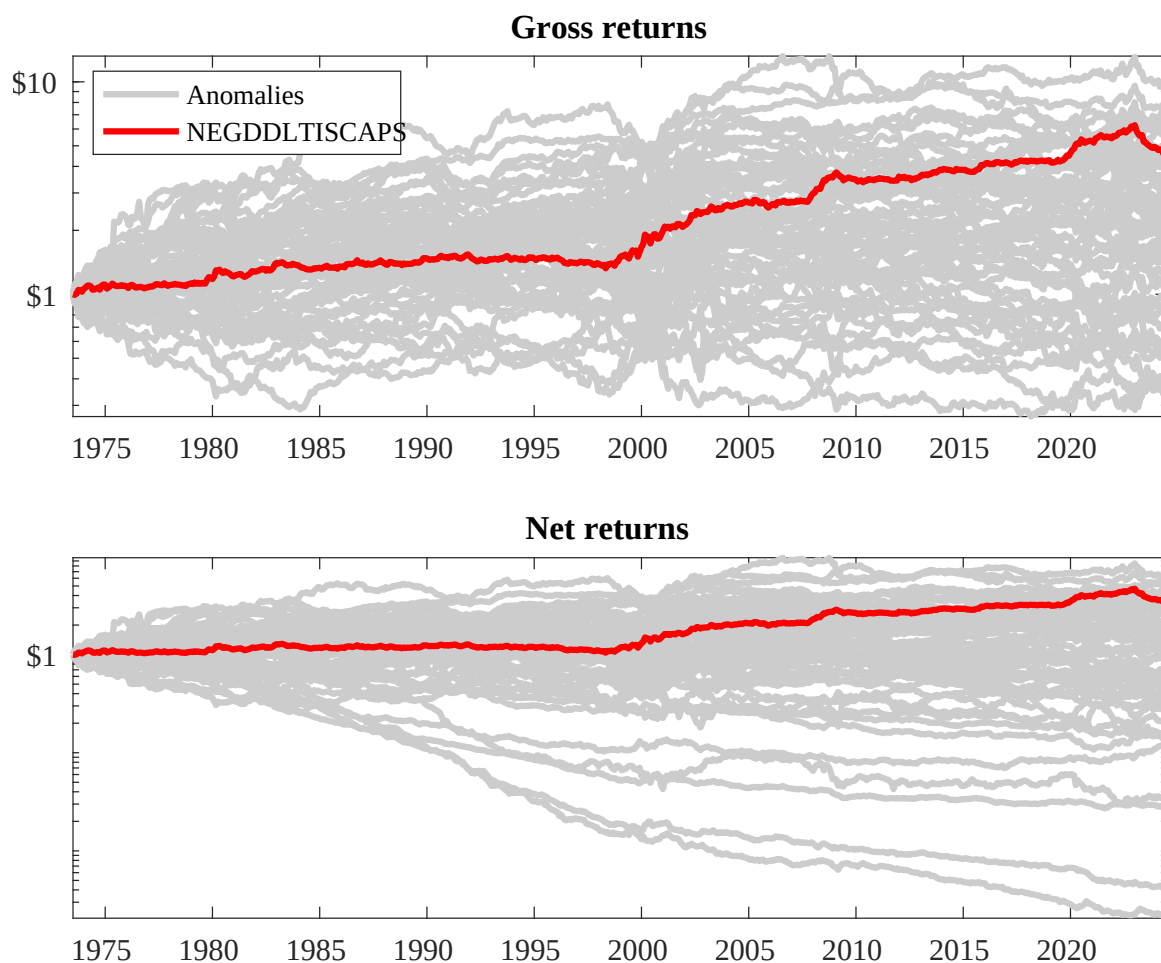
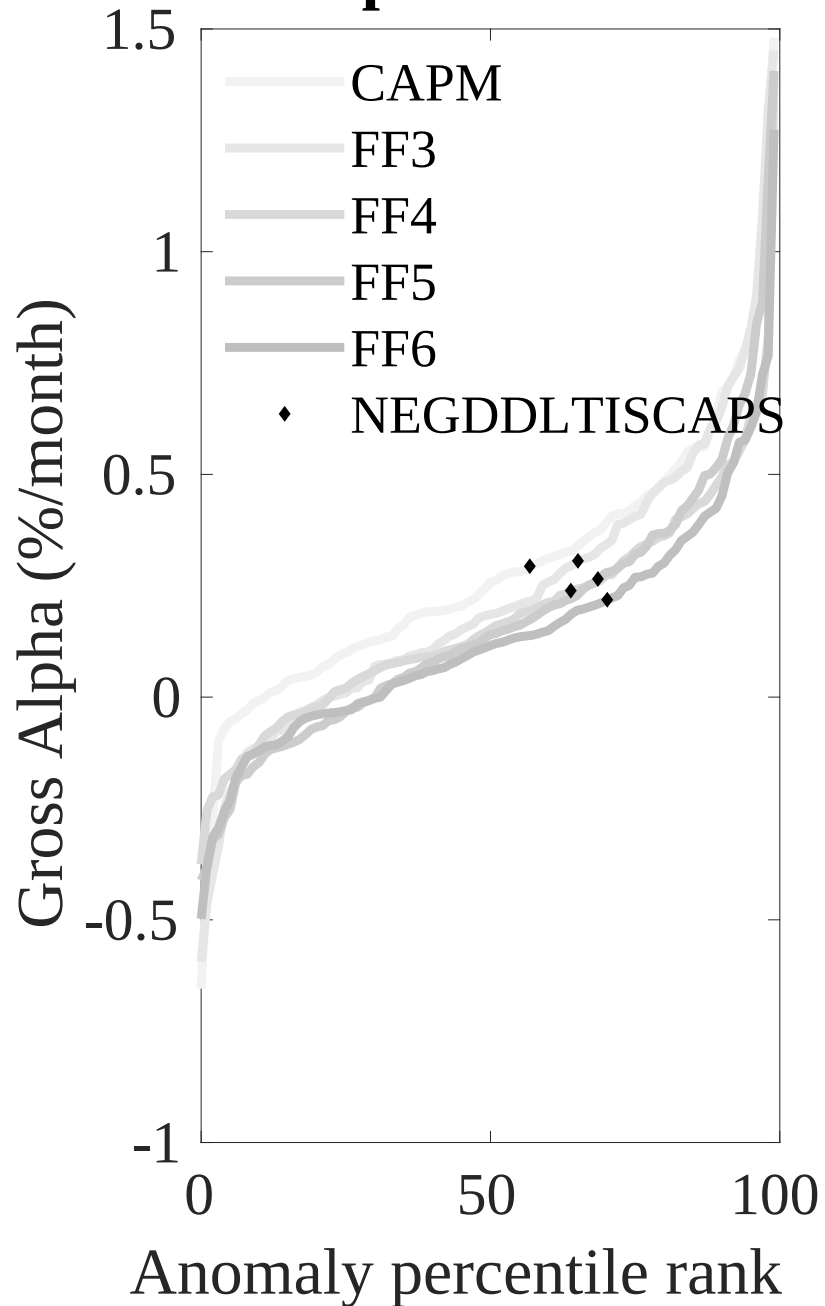


Figure 3: Dollar invested.

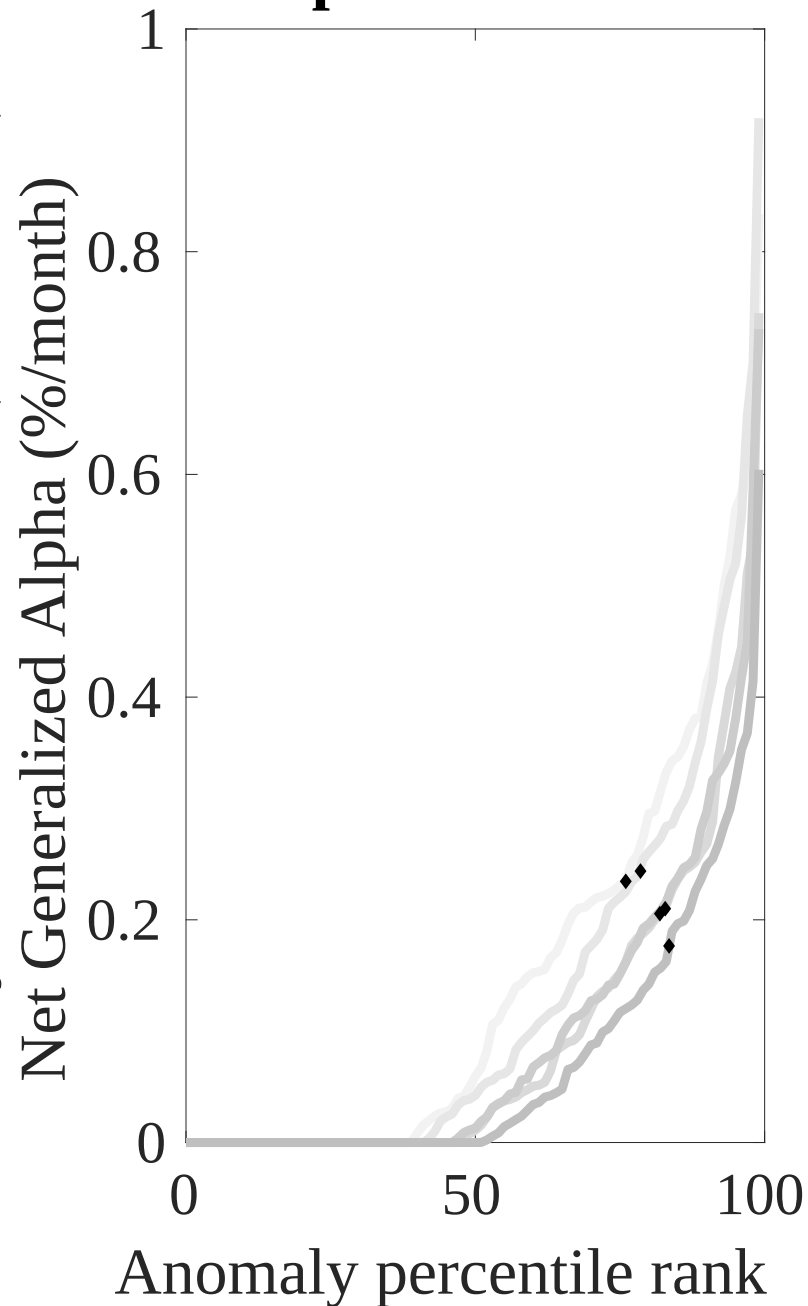
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EBP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



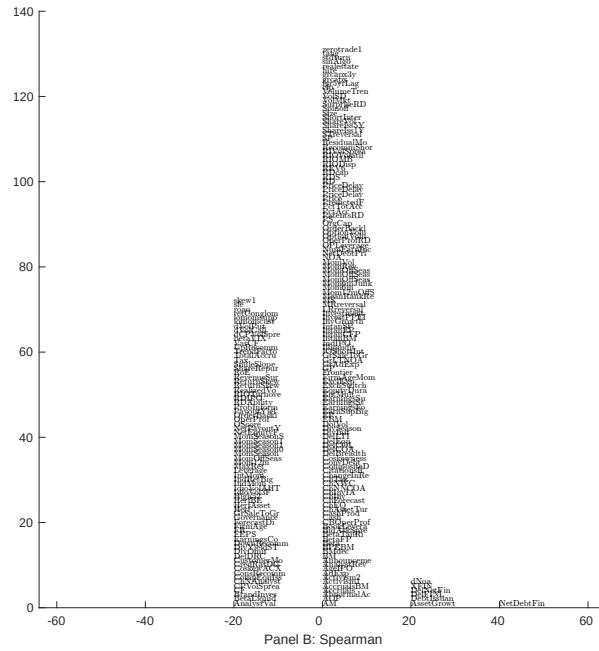
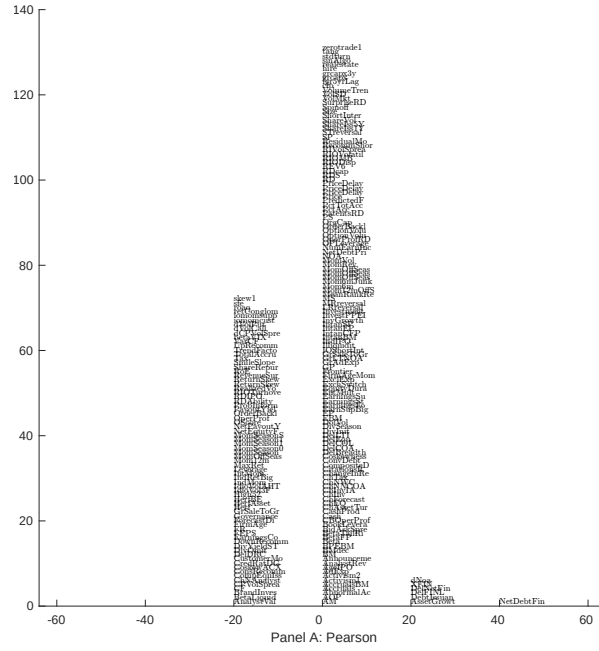


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EBP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

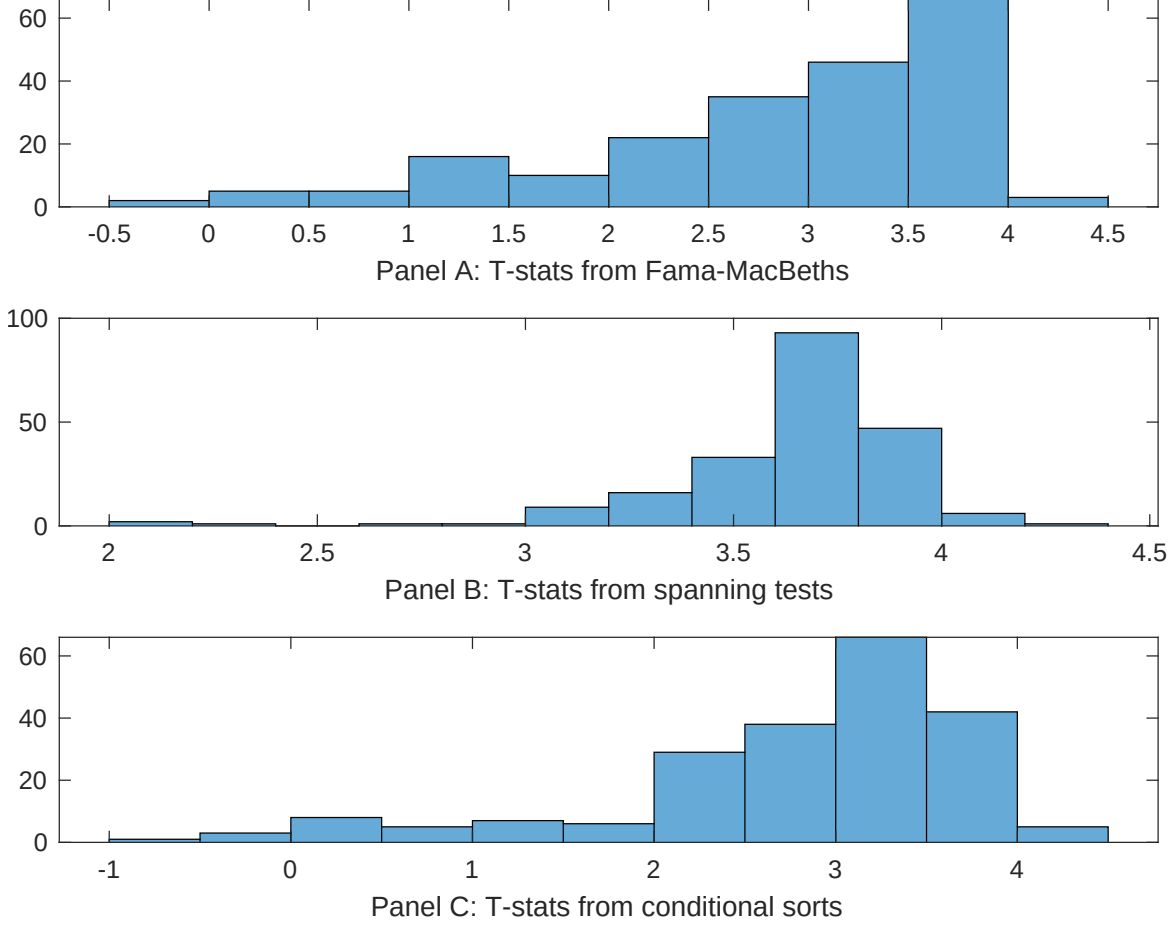


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EBP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBP}EBP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EBP. Stocks are finally grouped into five EBP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EBP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EBP}EBP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.28]	0.13 [5.25]	0.13 [5.22]	0.16 [6.27]	0.12 [4.92]	0.13 [5.19]	0.16 [6.47]
EBP	0.30 [0.53]	0.24 [0.43]	0.16 [2.48]	0.15 [3.11]	0.18 [3.42]	0.19 [3.69]	0.28 [0.40]
Anomaly 1	0.19 [9.45]						-0.18 [-0.04]
Anomaly 2		0.21 [8.96]					0.14 [2.73]
Anomaly 3			0.43 [7.25]				0.27 [4.08]
Anomaly 4				0.26 [5.99]			0.24 [4.60]
Anomaly 5					0.14 [4.38]		0.74 [1.80]
Anomaly 6						0.12 [1.27]	0.45 [0.38]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EBP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EBP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.58]	0.19 [2.62]	0.20 [2.80]	0.19 [2.67]	0.19 [2.59]	0.18 [2.48]	0.17 [2.38]
Anomaly 1	12.32 [2.98]						-2.36 [-0.42]
Anomaly 2		14.59 [3.67]					16.87 [3.14]
Anomaly 3			4.82 [1.69]				4.59 [1.64]
Anomaly 4				6.12 [2.15]			4.97 [1.76]
Anomaly 5					3.85 [1.32]		-3.84 [-1.17]
Anomaly 6						8.99 [3.41]	11.61 [3.79]
mkt	-0.68 [-0.41]	-0.94 [-0.56]	-1.06 [-0.63]	-1.10 [-0.65]	-0.68 [-0.40]	0.15 [0.09]	-0.01 [-0.00]
smb	5.64 [2.19]	5.62 [2.20]	7.43 [2.90]	5.77 [2.22]	7.64 [2.93]	5.96 [2.34]	3.28 [1.19]
hml	-11.15 [-3.44]	-11.38 [-3.53]	-12.12 [-3.73]	-11.55 [-3.55]	-10.98 [-3.30]	-5.37 [-1.43]	-3.75 [-1.00]
rmw	-6.86 [-2.08]	-7.36 [-2.23]	-5.06 [-1.52]	-5.24 [-1.59]	-4.49 [-1.31]	-3.36 [-1.00]	-4.47 [-1.29]
cma	18.92 [3.71]	19.26 [3.85]	18.75 [3.38]	22.46 [4.55]	21.79 [4.34]	22.96 [4.69]	16.08 [2.86]
umd	6.53 [3.83]	6.65 [3.96]	7.10 [4.18]	6.81 [3.98]	7.31 [4.32]	7.41 [4.45]	5.64 [3.26]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	11	12	10	11	10	12	14

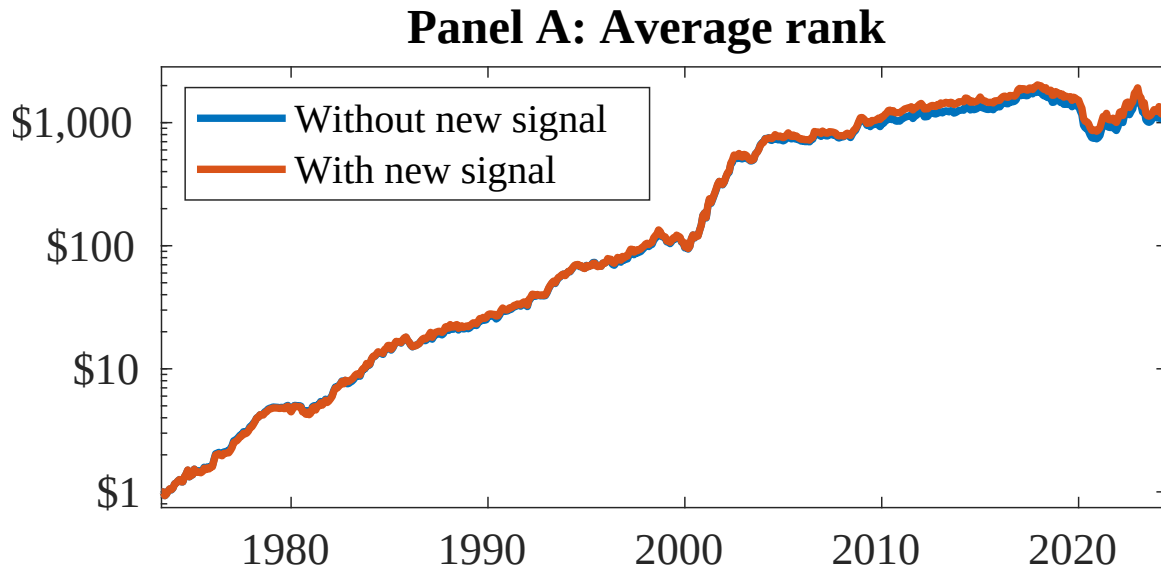


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EBP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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